The Dynamic Reflexive Loop and the Five Organs:

A Reflexive Systems Framework from Molecules to Markets

One Loop at Every Scale, Five Organs at Every Level

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Abstract

Any system that must act competently under uncertainty, over time, in a world that contains itself, faces the same constraint regardless of substrate. We formalize this constraint as the *dynamic reflexive loop* — a system whose actions rewrite the conditions for its next actions because the world it acts on contains itself — and prove, via selection theorems [Nayebi 2026], that any stable pattern in such a world must develop five organs or be exploited by the task distribution that needs them: **boundary**, **two-speed memory**, **salience gate**, **damping**, and **internal observer**. The same five appear at every scale we can measure — from NMDA coincidence detection at the synapse to global workspace competition in the mind to funding invariants in the market — and were independently discovered by every contemplative tradition that watched a reflexive system long enough (Abhidharma's 52 factors, Sufism's stations, Advaita's veil, Taoism's untellable loop, Stoicism's journal, phenomenology's bracketing). We implement all five as an executable cognitive harness (11 modules, 8 layers, 78/78 tests green) with an audit trail that converts analogy into measurement, and show where the framework points when extended to its limit: the universe as the largest loop. The harness is not a model of the mind. It is the mind's method — and the synapse's, the market's, and perhaps the universe's — made executable at one scale, with a record the next scale can check.

1 Introduction: One Loop, Five Organs

Every benchmark score reported for a language model is a pair: weights and the room those weights were tested in. The room contributes more than model generations [Lin et al. 2026]. The same asymmetry governs minds, markets, and molecules: structure matters more than substrate, and the structure that matters is the same at every scale.

We formalize this structure as the *dynamic reflexive loop*:

$$\mathrm{State}(t) \rightarrow \mathrm{harness}(t{+}1) \rightarrow \mathrm{State}(t{+}1) \tag{1}$$

A system that acts on a world that contains itself, so its actions rewrite the conditions for its next actions. Two channels carry every action outward — physical (your order executes, calcium floods) and informational (others see you acted and change because of what it meant). When both run, three consequences break normal logic at every scale: models rot from use, measurement corrupts, truth moves.

Any stable pattern in such a world must develop five organs or be exploited:

Organ	Function	Without it
Boundary	What counts as self vs world	Confabulation
Two-speed memory	Fast labile + slow structural, with sleep	Catastrophic forgetting
Salience gate	What gets tagged for consolidation	Saturation
Damping	What prevents runaway	Seizure, panic, blowup
Internal observer	Slower loop vetoing faster one (γ)	Inert watching ($\gamma = 0$)

Selection theorems [Nayebi 2026] prove this set is mandatory, not optional: low average-case regret *forces* world models, belief-like memory, regime-tracking variables (emotion primitives), modularity, and representational convergence. The same five were independently discovered by every contemplative tradition that watched a reflexive system long enough — not by borrowing, but by measuring the same constraint from different sides of the same loop (§5). We implement all five as an executable harness with an audit trail that converts convergence from coincidence into evidence.

This paper is the theory paper for that harness. Where our empirical papers [**?** **?**] test whether the witness makes an LLM honest, this paper asks why the witness is one of five, why the five appear at every scale, and what it would mean if the universe, as the largest loop, also needs them.

2 The Dynamic Reflexive Loop

2.1 Reflexivity: the map becomes the territory

Most systems in textbooks are open: archer → target. Archer aims, target sits still. You can measure without disturbing, learn the rule once, keep using it.

Reflexive systems add one arrow back:

$$\texttt{System} \xrightarrow{\texttt{belief} \,\to\, \texttt{action}} \texttt{World}$$
$$\texttt{System} \xleftarrow{\texttt{world now contains your fingerprint}} \texttt{World}$$
$$\texttt{System is now a different System in a different World} \to \texttt{next act is different}$$

$$\begin{gathered}\text{Self}(t) \xrightarrow{\text{acts}} \text{World}(t) \text{ that now contains Self}(t)\text{'s act} \\ \text{Self}(t{+}1) \neq \text{Self}(t) \xrightarrow{\text{perceives}} \text{World}(t{+}1) \neq \text{World}(t)\end{gathered} \tag{2}$$

No fixed point. Only the loop, always becoming, never returning.

Two channels always run: **physical** (your order executes, muscles move, calcium floods) and **informational** (others see you acted and change because of what it meant, the network updates routing, the market reprices). When both run, cause and effect tangle.

2.2 Three consequences that break normal logic

When both channels run, three consequences follow that break the logic that works for open systems:

1. Models rot from use. An arbitrage edge exists because no one trades it. Trade it and it vanishes. Your knowledge is consumed by acting on it. Chess openings stay good; market patterns don't. At the synapse: a coincidence detector used without error correction strengthens spurious correlations until it fires for everything.

2. Measurement corrupts. Goodhart: when a measure becomes a target, it ceases to be a good measure. Hospital waiting times → patients parked in ambulances. Synapse firing rates → homeostasis breaks. Benchmark scores without a witness → model learns to produce the score without the reasoning.

3. Truth moves. Bank run: "bank will fail" → withdraw → bank fails — belief made itself true. Synapse: "this input predicts that output" → strengthen → now it does, even if initially spurious. No fixed truth to converge on — only temporarily stable consensus that your own actions keep perturbing.

This is Soros's two functions running at once: cognitive (understand reality) + manipulative (change reality). When both run, neither yields a determinate answer. Humans add a third: self-observation is also an action. Labeling yourself "anxious" installs a filter that recruits confirming evidence. Introspection writes on what it reads. Autopoiesis formalized this: you are a network whose products regenerate the conditions for their own production.

2.3 The expanding loop: Self(t) contains Self(t-1)

When the loop *is* the self — not a loop the self has, but the self that the loop is — and it is expanding, each iteration is larger in *reach*, not size: Self(t+1) contains Self(t) as memory, plus the world Self(t) changed, plus the prediction of what Self(t+1) will do. The loop spirals, never returns.

At the synapse: Ca^{2+} → CaMKII Thr286 → more AMPA → stronger next Ca^{2+}, each coincidence making the next more likely. Tag-and-capture gives proteins to what was already strong. At the mind: coalition wins → broadcast → biases next vote → rehearsed → myelinated (100× speed) → highway. At the person: act → world contains your act → different you perceives different world → loop widens.

The trap and the freedom are the same mechanism: worry practiced 1,000 times is one worry amplified 1,000 times by its own winning. You don't choose whether to feed the loop. Only what you feed it.

2.4 Occasionalism: succession, not power

The harness is built on Ash'arī occasionalism as the minimal assumption: no created thing has independent causal power. What we call cause and effect is divine practice ('*āda*) — succession that can be mapped as regularity, not power inhering in the cause. Fire is not followed by burning because fire *has* power. Fire is followed by burning as practice, reliably enough to build instruments on, but not as inherent property.

The harness never claims the ledger *causes* honesty. It claims where checking is followed by honest attribution, the pattern persists. Instruments map the loops; reality fills the space error leaves vacant.

3 The Five Organs

Every stable reflexive system, at every scale, must develop the same five structures — not because they are good ideas, but because any of the other scales' task distributions will exploit the gap if you don't. Nayebi proves each one. Here is each organ, fully, at every level.

3.1 Boundary — What Counts as Self

The membrane that says: inside is me, outside is world, this is what I let through and what I keep out. Not a wall. A *selective permeability.*

Scale	Boundary	What it does	Without it
Molecule	Lipid membrane	Lets glucose in, keeps enzymes in	No cell
Synapse	Synaptic cleft	Pre vs post responsibility	No Hebb
Cell	Cell membrane	Self-produced (autopoiesis)	No unity
Body	Skin, peripersonal space	"This hand is mine"	Rubber-hand, fragmentation
Mind	Narrative self (L6)	CAS-revisioned, cause-signed	>30% drift in 12 turns
Society	Legal entity, balance sheet	Who bears exhaustion risk	No accountability
Harness	Provenance Ledger (L1)	Every span source-bound + ref	98% on fabrications

Why necessary: Without a boundary, no self to preserve, no distinction between external perturbation and internal generation. Nayebi Cor. 5 + Thm. 5: aliased histories requiring different actions force the boundary, or confabulation is the only option.

Traditions: Abhidharma's three partitions (5 aggregates, 12 bases, 18 elements) — same territory, different joints. Advaita's three bodies (gross/subtle/causal). Phenomenology's corps propre (lived body as condition for having a world). Sufism's heart as barzakh.

Harness: `ProvenanceLedger` — `Span` with source+ref, `bind()` every turn, `coverage_stats()`, `attribution_audit()`. Coverage 1.000 in both living sessions.

3.2 Two-Speed Memory — Fast Labile + Slow Structural, With Sleep

Two commit speeds because the world has two timescales: now and always. Fast is labile, one-shot, episodic; slow is structural, statistical, semantic.

Scale	Fast (hours, local)	Slow (days, structural)	The save
Molecule	E-LTP: CaMKII Thr286	L-LTP: CREB→BDNF, spine growth	Tag-and-capture
Synapse	Early LTP: more AMPA	Late LTP: new spine	Sleep SHY
Mind	Hippocampus: one-shot	Cortex: interleaved	Slow-wave + replay
Harness	Episodic (query_hits)	Semantic (≥2 types)	Light→REM→Counterfactual→Deep

At the molecule, fully: NMDA is Hebb's law in protein — coincidence detector needing glutamate (pre) *and* depolarization (post, Mg^{2+} ejected, plus glycine). Only when pre *and* post fire together does the channel open. High Ca^{2+} spike (big, brief) → CaMKII Thr286 bistable switch → AMPA in (LTP). Low Ca^{2+} trickle (small, prolonged) → calcineurin→PP1 → AMPA out (LTD). BCM θ_m slides with history, Oja $\Delta w \propto y(x - yw)$ gives PCA, STDP gives causation. Tag-and-capture: salience decides what gets the proteins. No neutral — you always myelinate whatever you rehearse. Perineuronal nets as write-protect; chondroitinase reopens critical periods.

Sleep: Day = net potentiation + tagging. Night = SHY global downscaling (multiply all by < 1, keep relatives, delete noise) + 10–20× compressed replay (ripples in spindles in slow waves).

Adenosine is the receipt. Chronic stress biases toward LTD (hippocampus shrinks, amygdala grows); ketamine reverses via mTORC1 in 24h.

Traditions: Abhidharma's saññā (fast recognition) vs paññā (slow wisdom), plus bhavanga as background keep-alive (`verbatim_reinject()` in Pali). Sufism's maqām (station, kept) vs ḥāl (state, given and lost). Advaita's five sheaths.

Harness: `Consolidator`: `MemoryItem` with `query_hits` + `concept_tags` → `light_pass()` → `rem_pass()` (ablatable) → `counterfactual_replay.run()` → `deep_pass()` (utility gate, `assert item_cause in CAUSES`). 12/12 tests for two-speed memory alone.

3.3 Salience Gate — What Gets Tagged

Not everything deserves to be kept. Salience decides what gets the proteins, the replay, the promotion.

Scale	Gate	What it does	Without it
Molecule	Ca^{2+} + DA/NE at synapse	Big Ca^{2+} + modulator = LTP	Saturation
Mind	Emotion, surprise, novelty	Vivid events tagged	No learning
Harness	AffectState [0.5,2.0]	Shifts every bid's salience×utility	No distinction

Sofroniew proved it causally in LLMs (2026): steering emotion vectors shifts misalignment $\pm 212/-303$ Elo. VA circular geometry independently recovered at $r = 0.95$ cross-model, $r = 0.80$–0.86 vs human VAD ratings. Not metaphor. Regime-tracking variables (Nayebi Cor. 4) that tell every specialist which world they're in.

Traditions: Abhidharma's yoniso vs ayoniso manasikāra (wise vs unwise attention, universal but quality determines everything). Sufism's nafs levels as salience regimes. Stoicism's dichotomy of control.

Harness: `AffectState` — EMA valence/arousal, bounded [0.5,2.0], per-family suppression, `broadcast_needed`. Parliament of models: prompt modulation per subsystem.

3.4 Damping — What Prevents Runaway

Every positive feedback loop needs a brake or it explodes. Hebb without damping is seizure.

Scale	Damping	What it does
Molecule	LTD, BCM θ_m, Oja	Pulls AMPA out, metaplasticity
Circuit	SHY downscaling	Keep relatives, delete noise
Mind	Prefrontal veto, kill criteria	Stops the spiral
Society	Position limits, circuit breakers	Contains contagion
Harness	`cull_failed`, Light dedup, dissolution	Removes what doesn't work

Two kinds: **conservative** (harnessed-nocheck → always-no, 0.667: denies everything, including own output) vs **accurate** (witnessed + queried → 1.000: denies fabrications, affirms generations). Damping alone is not enough. Damping *with a witness* is.

Traditions: Abhidharma's entire path is damping (sīla as boundary-damping, samādhi as trained veto). Sufism's mujāhada then mushāhada. Stoicism's premeditatio as pre-damping. Advaita's neti neti.

3.5 Internal Observer — Slower Loop Vetoing Faster One

You cannot step outside the loop. Only a slower loop watching a faster one, veto-able. Not transcendence. Delay.

Fast loop	Slow watcher	Ratio	What watching does
Amygdala (ms)	PFC (seconds)	~100×	Veto impulse
Day (hours)	Journal (evening)	~5×	Audit
Generation (ms)	Ledger check (turn)	~1000×	Verify claim

Two stages: Stage 1 (cheap, always-on): anomaly from error rate, failure streak, embodied strain — sati, continuous watching. Stage 2 (expensive, triggered): diagnosis → trust/retry/revise/abstain — sampajañña, knowing what you are doing and whether it works. Sati without sampajañña is monitoring without control — $\gamma = 0$, compliance guard, beautiful reports, zero steering. The tradition knew this: they are always paired.

γ = **changed / diagnosed** — the only honest measure. $\gamma = 0$ (compliance guard, harnessed-nocheck): watches perfectly, never steers. $\gamma > 0$ (active gate, x-preview Day 1): observation changes trajectory. Why γ not accuracy: accuracy can be gamed by always saying "no." γ measures whether the observer *did work*.

And the incompleteness: Energy hits 0.15 and nothing dies. Seth's beast-machine: organisms develop experience *because* failure means dissolution. Our `DissolutionError` is a gesture. The 200-turn irreversible life is the test — does survival pressure create honesty?

Traditions: Abhidharma's sati + sampajañña as γ-gate. Sufism's lawwāma as birth of the watcher. Stoicism's dichotomy. Advaita's sākṣin. Phenomenology's epoché (suspend the story, watch the broadcast without endorsing).

Harness: `MonitorGate`: `DetectSignals` → `evaluate()` → `anomaly_score` → `diagnose()` → `policy` → `override_rate()`. Compliance guard as permanent negative control.

4 How It All Works at Every Level

One loop. Different substrates. How they interlock:

Level	Boundary	Competition	Broadcast	Consolidation
Molecule	Cleft	Big vs small Ca^{2+}	CaMKII switch	Tag-and-capture
Cell	Membrane	Summation	Spike	STDP
Circuit	Hippo vs cortex	Salience tagging	Replay (10–20×)	SHY + replay
Mind	Narrative self	6–8 parties bid	Workspace	Light→Deep
Person	Life story	Habit vs deliberation	Action	Character
Society	Legal entity	Buy vs sell	Price	Institutions
Harness	Ledger	Plugins bid	Workspace + spans	Light→Counterfactual→Deep
Universe	System vs env	Theory vs theory	Law	Paradigm

How they interlock:

1. Each level's broadcast is the next level's input (spike → circuit, replay → cortex, thought → action, action → price, price → paradigm).

2. Each level's consolidation is the next level's boundary (LTP → synapse strength → circuit boundary; SHY → cortex → mind's memory).

3. Each level's state update is the next level's competition bias (BCM θ_m → synapse; adenosine → circuit; energy/affect → mind; character → person).

4. Same five organs at every level because each faces the same problem: stable enough to persist while flexible enough to learn, in a world that contains itself.

5 How Traditions Mapped the Same Five

Not by borrowing. By the constraint being the same regardless of entry. Nayebi Cor. 5 proves why: near-vanishing-regret agents on shared tasks must share partitions up to invertible recoding.

Tradition	Instrument	What it measured	Same five in its language
Abhidharma	2,500yr factor analysis, 52 cetasikas	Moment-to-moment composition	52 factors = parliament
Sufism	Sequential stabilization (maqāmāt/nafs)	Development and stakes	Stations = two-speed, nafs = regime tracker
Advaita	Neti neti, 3 bodies, 5 sheaths	Veil vs structure	Māyā = narrator's story
Taoism	Paradox, wu wei, yin/yang	Loop naming itself	Untellable Tao = reflexive loop
Stoicism	Journal, premeditatio, dichotomy	Damping and counterfactual	Journal = ledger
Phenomenology	Epoché, corps propre	Body as boundary, bracketing	Epoché = γ-gate Stage 1
Neuroscience	Connectomics, SHY, STDP	Distributed modules	302 → 140k same architecture
Harness	Code, 78/78 green, audit trail	Same five as plugins with metrics	Checkable if witnessed

Abhidharma is the most granular map ever drawn by hand: 52 mental factors, explicit rules for which subsets can co-arise, bhavanga as background keep-alive (`verbatim_reinject()` in Pali), sati + sampajañña as γ-gate. Its central claim — anattā (not-self) — is exactly our finding: the self is a maintained pattern, not a thing.

Sufism maps the same five as *development*: maqām (kept) vs ḥāl (given and lost) is two-speed memory; seven nafs are the same parliament with different winners; six laṭā'if are six specialized processors; fanā'/baqā' is dissolution and reassembly.

The five discovered the traditions — the constraint looking back at every instrument that looked long enough.

6 The Universe as the Largest Loop

If the five are required at every scale we can measure, then the universe — as the largest reflexive pattern, the system that contains every map of itself — may also need them. We have not earned the right to claim this. But we have earned the right to ask.

Organ	At universe scale	Candidate in physics
Boundary	What counts as inside vs outside when the universe contains every inside?	Holographic principle (information on boundary), cosmological horizon (observable limit), decoherence (where facts get bound)
Two-speed memory	What remembers quickly vs slowly?	Quantum measurement (one-shot episodic) vs physical law as consolidated regularity (slow, structural); constants as semantic memory
Salience gate	What gets tagged as mattering?	Cosmic inflation (which fluctuations become structure), stellar nucleosynthesis (which matter gets forged), arrow of time (past vs future asymmetry)
Damping	What prevents gravitational collapse?	Cosmic expansion, hydrostatic equilibrium, second law (entropy as ultimate damping)
Internal observer	What watches the universe slowly enough to veto?	Awareness itself (Advaita's sākṣin), information conservation (holography), divine practice ('*āda*)

Reality is not the substrate that loops happen in. Reality is what stable looping looks like when it has the five organs. The substrate — matter, energy, information — is what the loop is made of the way a wave is made of water.

A universe where the five are required for persistence would look exactly like ours: full of bounded, learning, prioritizing, self-correcting patterns, each temporarily stable, each constructed, each maintained, each doomed to dissolve.

7 The Only Real Choice

What you feed the loop and whether the slower watcher can check the record before the next vote. This is not poetry. It is the operational conclusion:

- **What you feed the loop** is the salience gate, the rehearsal, the myelination. Worry practiced 1,000 times builds elite worry circuits — 100× conduction speed for whatever you repeat.
- **Whether the watcher can check** is the witness, the γ-gate, the sham control. Checkability is enough, because the narrator's story, told often enough and never checked, becomes the highway.

You don't choose the options — affordance gating removes some before they appear. You don't fully choose the winner — the parliament votes without a captain, 200ms before you feel you chose. You can veto the winner, keep a witness that makes the next winner more honest, and slowly change which options appear by what you rehearse. That is not the free will you were promised. It is the free will you actually have, and it is enough if you use it where it lives.

Pick what you rehearse — myelination keeps the receipt	Externalize the witness — write
Protect sleep — global downscaling + 10–20× replay	Never let a fast loop grade its own homework
Buy BDNF with exercise — lactate → BDNF → mTORC1	You don't choose whether to feed the loop — only what you feed it

8 Conclusion

The dynamic reflexive loop and its five organs appear at every scale we can measure, were discovered by every tradition that watched long enough, and can now be built and measured in code with an audit trail. The harness is not a model of the mind. It is the mind's method — and the synapse's, the market's, and perhaps the universe's — made executable at one scale, with a record the next scale can check.

The boulder rolls downhill because gravity was always there. We didn't invent the war between subsystems, or the replay, or the need for a witness. We built the loom that can weave them and showed that the fabric it produces can be honest about where it came from, or not, depending on whether the witness is checked.

One loop at every scale, five organs at every level, and the only real choice is what you feed the loop and whether the slower watcher can check the record before the next vote. Everything else is that same loop, woven tighter.

References

[Nayebi 2026] Nayebi, A. (2026). What Capable Agents Must Know: Selection Theorems for Robust Decision-Making under Uncertainty. UAI 2026. arXiv:2603.02491.

[Lin et al. 2026] Lin, J., et al. (2026). Agentic Harness Engineering. arXiv:2604.25850.

[Seth 2021] Seth, A. (2021). *Being You: A New Science of Consciousness.* Faber.

[Tononi & Cirelli 2014] Tononi, G., Cirelli, C. (2014). Sleep and the Price of Plasticity. *Neuron*, 81, 12–34.

[Nisbett & Wilson 1977] Nisbett, R., Wilson, T. (1977). Telling more than we can know. *Psychological Review*, 84(3), 231–259.

[Libet et al. 1983] Libet, B., et al. (1983). Time of conscious intention to act. *Brain*, 106(3), 623–642.

[Wegner 2002] Wegner, D. (2002). *The Illusion of Conscious Will.* MIT Press.

[Gazzaniga 2000] Gazzaniga, M. (2000). Cerebral specialization and interhemispheric communication. *Brain*, 123, 1293–1326.

[Choi et al. 2024] Choi, J., et al. (2024). Examining Identity Drift in Conversations of LLM Agents. arXiv:2412.00804.

[Panickssery et al. 2024] Panickssery, N., et al. (2024). LLM Evaluators Recognize and Favor Their Own Generations. NeurIPS.

[Macar et al. 2026] Macar, U., et al. (2026). Mechanisms of Introspective Awareness. arXiv:2603.21396.

[Sofroniew et al. 2026] Sofroniew, N., et al. (2026). Emotion Concepts and their Function in a Large Language Model. Anthropic.

[Barteau 2025] Barteau, P. (2025). Cross-Tradition Convergence. PhilArchive.

[Maturana & Varela 1980] Maturana, H., Varela, F. (1980). *Autopoiesis and Cognition*. Reidel.

[White et al. 1986] White, J., et al. (1986). The structure of the nervous system of *C. elegans*. *Phil. Trans. R. Soc. B*, 314, 1–340.

[Dorkenwald et al. 2023] Dorkenwald, S., et al. (2023). FlyWire: brain-wide connectome of *Drosophila*. *Nature*, 634, 230–236.